

Phase I/II Statistical Process Monitoring

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統計專論 (III) Final Report

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Project Objective

- Dataset: UCI SECOM
- Objective: Construct SPC workflow

Preprocessing → Phase I → Control chart design → Phase II

- Main comparison: Shewhart, EWMA, CUSUM, and CPD charts
- Final:
 1. Which chart signals first ?
 2. Whether the pattern suggests mean shift, variance shift, or structural change ?

Selected Variable and Preprocessing

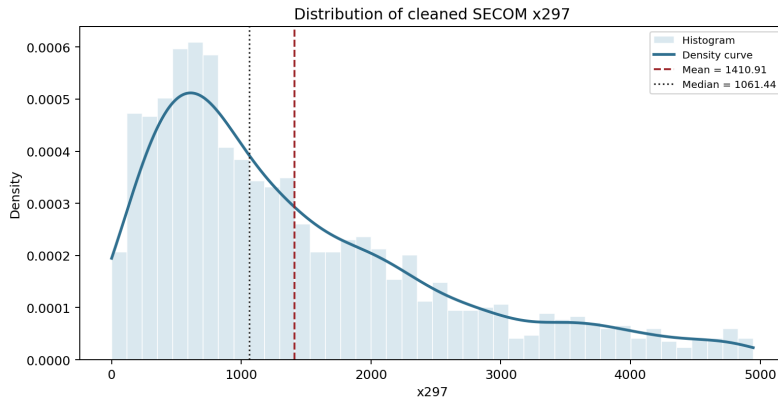
Variable

Selected process variable: x_{297}

- Reason: Low missing, High distinct
- Samples: 1567
- Missing samples removed: 2
- Outlier: 1.5 IQR filter after dropping missing values
- Outliers removed: 128
- Cleaned samples: **1437**

Variable Summary

Mean	SD	Min	Q1	Median	Q3	Max
1410.91	1126.51	0	556.99	1061.44	1995.39	4945.92



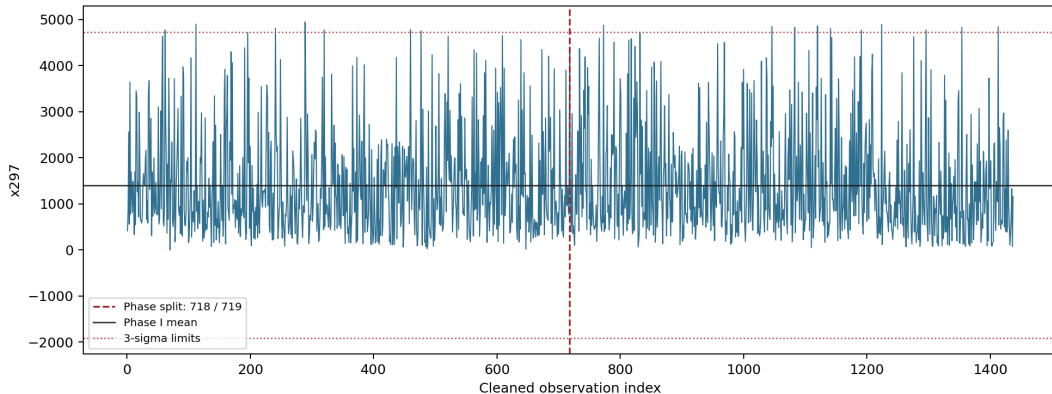
Phase I / Phase II Split

$$m = \lfloor 0.5n \rfloor = \lfloor 0.5(1437) \rfloor = 718$$

Phase	samples	Mean / SD
Phase I	718	1403.64 / 1105.53
Phase II	719	1418.18 / 1147.80

- Phase I is used as the in-control reference sample
- Phase II is monitored sequentially using the chart limits estimated from Phase I

Cleaned Series and Phase Split



Phase I RS/P Diagnosis

Recursive Segmentation + Permutation Calibration

$$p_{\text{level}} = 0.8675, \quad p_{\text{scale}} = 0.8325$$

- Phase I shows no significant evidence of level or scale change
- Phase I data can be regarded as stable

Chart Design

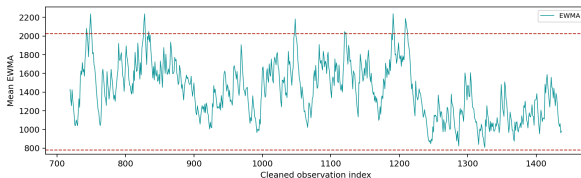
Method	Target	Main setting
Individuals Shewhart	Mean	3-sigma limits
Mean EWMA	Mean	$\lambda = 0.1, \rho = 2.454$
Variance EWMA	Variance	$\lambda = 0.1, \rho_U = 2.595, \rho_L = 1.580$
Adaptive EWMA	Mean	$\eta_1(e_n): \mu_1 = 1, \mu_2 = 4, ARL_0 = 100,$ $h = 0.7874, \lambda = 0.1813, u = 2.5752$
Mean CUSUM	Mean	$k = 0.5, h = 4.095$
Variance CUSUM	Variance	$k_U = 1.848, h_U = 7.416; k_L = 0.462, h_L = -2.446$
CPD charts	Mean / variance / joint	$\alpha = 0.005$

Mean-Shift Monitoring Results (Main Mean Charts)

Mean EWMA

First signal: 743

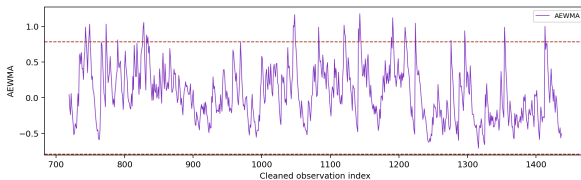
Signal count: 17



Adaptive EWMA

First signal: 743

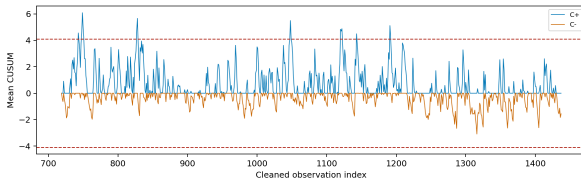
Signal count: 32



Mean CUSUM

First signal: 743

Signal count: 13

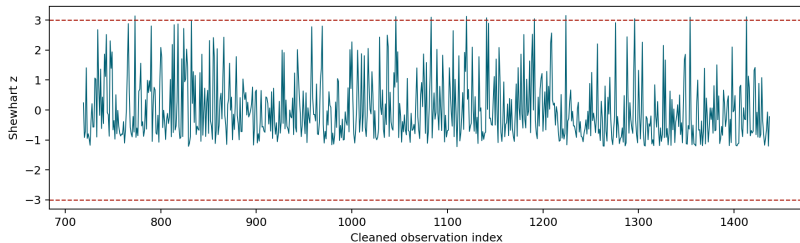


Mean-Shift Monitoring Results (Supplementary Charts)

Individuals Shewhart

First signal: 773

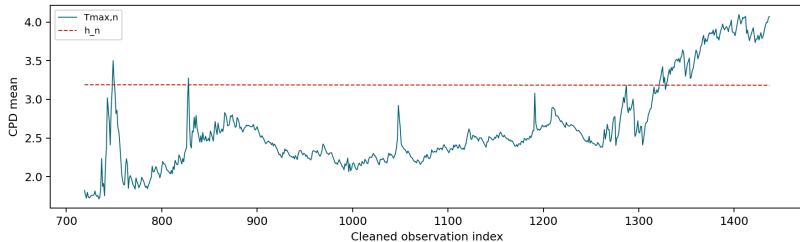
Signal count: 10



CPD Mean

First signal: 749

Signal count: 118

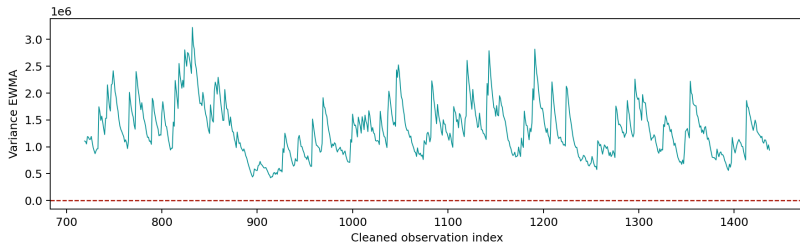


Variance-Shift Monitoring Results (Target: Variance)

Variance EWMA

First signal: 719

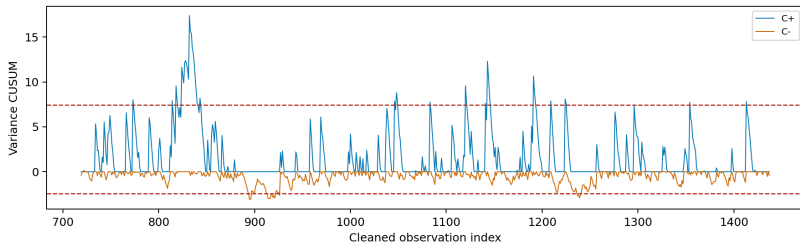
Signal count: 719



Variance CUSUM

First signal: 773

Signal count: 59

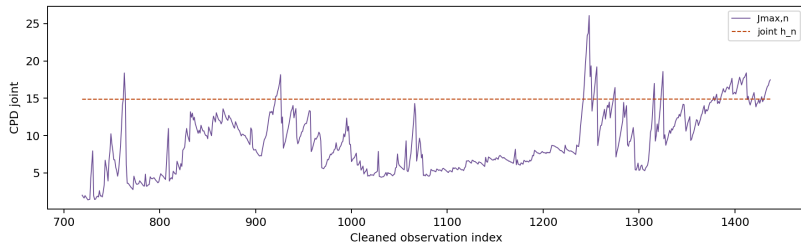


CPD and Joint Monitoring Results

CPD joint

First signal: 762

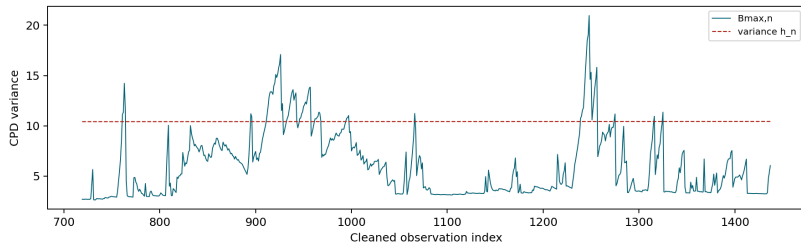
Signal count: 71



CPD variance

First signal: 761

Signal count: 78



CPD Post-Signal Interpretation

- The strongest CPD mean signal estimates:

$$\hat{r} \approx 1212$$

- The strongest CPD variance and joint signals estimate:

$$\hat{r} \approx 1229$$

Conclusion

- ① The selected SECOM variable x_{297} is usable
- ② Phase I RS/P results support using the first 718 samples as the IC reference
- ③ The variance EWMA is very sensitive
- ④ CPD charts suggest structural changes around indices 1212–1229